

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

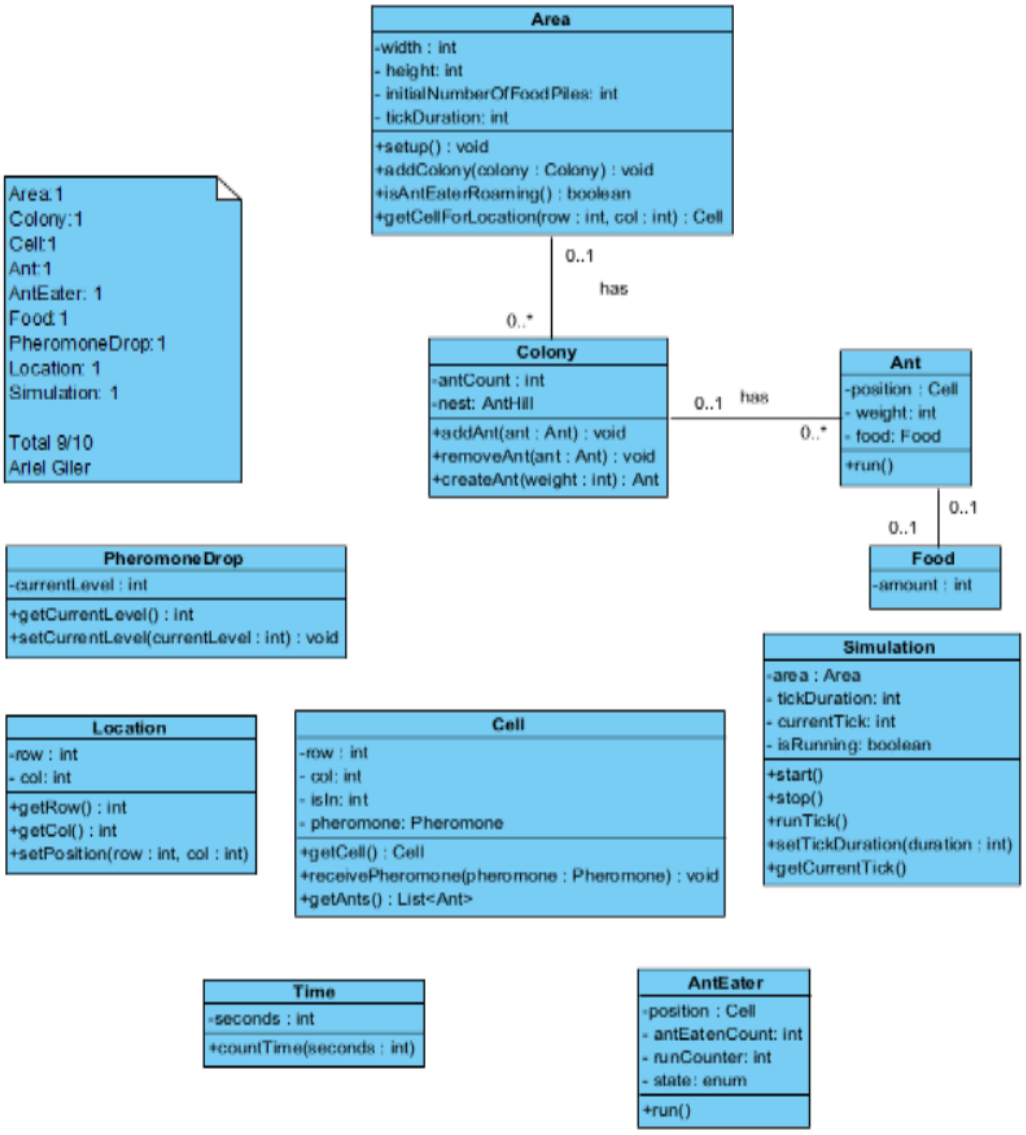
Object Oriented Programming

36863

WorkShop11-FindingObjectsAntsAndAntEaters

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Area
-width : int
- height: int
- initialNumberOfFoodPiles: int
- tickDuration: int
+setup() : void
+addColony(colony : Colony) : void
+isAntEaterRoaming() : boolean
+getCellForLocation(row : int, col : int) : Cell
0..1
has
0..*
Area:1
Colony:1
Cell:1
Ant:1
AntEater: 1
Food:1
PheromoneDrop:1
Location: 1
Simulation: 1
Total 9/10
Ariel Giler
Colony
-antCount : int
-nest: AntHill
+addAnt(ant : Ant) : void
+removeAnt(ant : Ant) : void
+createAnt(weight : int) : Ant
0..1
has
0..*
Ant
-position : Cell
- weight: int
- food: Food
+run()
0..1
0..1
Food
-amount : int
PheromoneDrop
-currentLevel : int
+getCurrentLevel() : int
+setCurrentLevel(currentLevel : int) : void
Simulation
-area : Area
- tickDuration: int
- currentTick: int
- isRunning: boolean
+start()
+stop()
+runTick()
+setTickDuration(duration : int)
+getCurrentTick()
Location
-row : int
- col: int
+getRow() : int
+getCol() : int
+setPosition(row : int, col : int)
Cell
-row : int
- col: int
- isIn: int
- pheromone: Pheromone
+getCell() : Cell
+receivePheromone(pheromone : Pheromone) : void
+getAnts() : List<Ant>
Time
-seconds : int
+countTime(seconds : int)
AntEater
-position : Cell
- antEatenCount: int
- runCounter: int
- state: enum
+run()